Next Permutation (medium)

Problem Statement

Given an array of integers `nums`, return the next permutation of `nums.`

A `permutation` of the array is an `arrangement` of its elements in the `linear order` or `particular sequence`.

- For example, permutations of the `[2, 1, 3]` array are `[3,2,1], [3,1,2], [1,2,3], [1,3,2], [2, 1, 3]`, and `[2, 3, 1]`.

A `next permutation` of the array is the `next smallest lexicographically greater` permutation of array elements.

- For example, next permutation of `[2, 1, 3]` is `[2, 3, 1]`, which is smallest lexicographically greater permutation.

If `no higher permutation` is possible, the array should be rearranged into its `lowest possible order` (ascending).

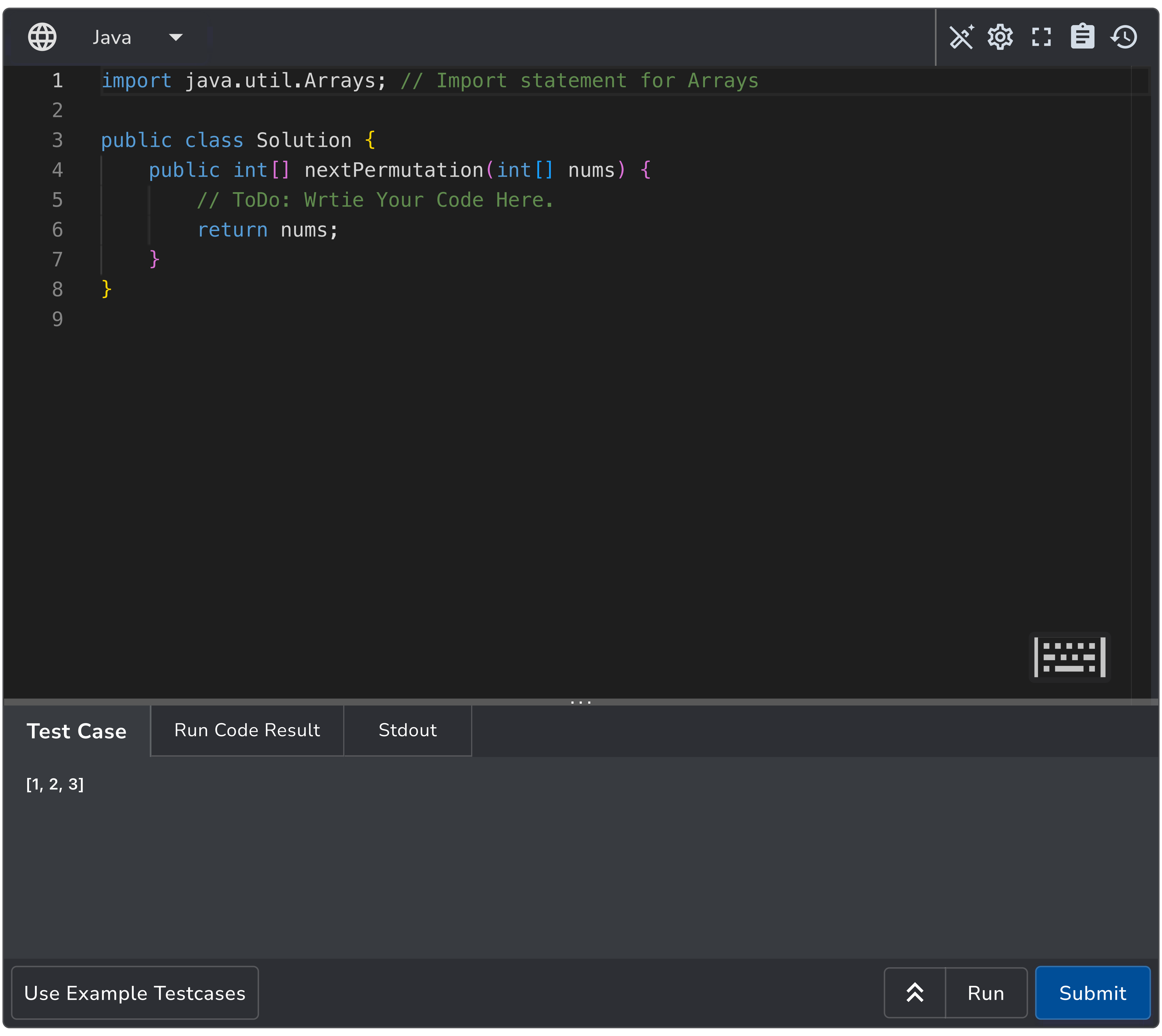
- For example, next permutation for `[3, 2, 1]` is `[1, 2, 3]` as `[3, 2, 1]` doesn't have lexicographically larger permutation.

Examples

- **Example 1**:
  - **Input**: `[1, 3, 5, 4, 2]`
  - **Expected Output**: `[1, 4, 2, 3, 5]`
  - **Justification**: The next permutation is obtained by swapping `3` with the next higher number (`4`) and then sorting the numbers after `4` in ascending order.
- **Example 2**:
  - **Input**: `[2, 3, 6, 5, 4, 1]`
  - **Expected Output**: `[2, 4, 1, 3, 5, 6]`
  - **Justification**: The next permutation is found by swapping `3` with the next higher number (`4`) in the sequence following it and then sorting the numbers after `4` in ascending order.
- **Example 3**:
  - **Input**: `[1, 1, 5]`
  - **Expected Output**: `[1, 5, 1]`
  - **Justification**: The next immediate permutation after `[1, 1, 5]` is `[1, 5, 1]`.

Try it yourself

Try solving this question here:



```java
import java.util.Arrays; // Import statement for Arrays

public class Solution {
    public int[] nextPermutation(int[] nums) {
        // ToDo: Wrtie Your Code Here.
        return nums;
    }
}

```

[1, 2, 3]

← Previous
Solution: Subarray Sum Equals K
Next →
Solution: Next Permutation

☐ Mark as Completed